A Spherical Chiplet Fabric with Geodesic Topology, Planet-Style Coordinates, and Reorientable Frame Semantics for Large-Scale AI and HPC Workloads

Finite-Buffer, Inference and RTL Prototypes

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Abstract

We investigate a geodesic chiplet interconnect and revise the evidence for a proposed accelerator built around it. A corrected Python store-and-forward baseline uses chronological hop events, independent directed links and infinite output queues. Separate executable prototypes add bounded flit buffers and credits, a tiny causal NumPy MoE decoder with a KV cache, a bounded DATA/ACK recovery protocol, and a synthesizable local word fabric. Corrected experiments compare equal-size sphere and torus graphs, declare their bandwidth budgets, and separate isolated probes from contending traffic. A 40-profile study with five paired seeds covers synthetic mixture-of-experts (MoE) dispatch, a torus-local stencil and static single-link removals. Matching aggregate link bandwidth gives median sphere/torus MoE p99 ratios of 1.226 at 42 ranks and 1.158 at 162 ranks. Equal per-link rates reverse that ordering, with ratios of 0.834 and 0.779, while supplying the sphere with more links and aggregate bandwidth. These are model-specific results, not general hardware rankings. An analytical audit distinguishes 30 TB/s of assumed local HBM bandwidth from at most 4.8 TB/s through the historical memory tiles' fabric ports. This motivates a four-island compute-near-HBM architecture with explicit memory ownership and a hierarchical network. The decoder exercises logical ownership on four islands and agrees with an independent full-prefix numerical reference; all arithmetic runs in one CPU process with synthetic weights. A fixed-tree channel-dependency certificate covers the bounded network's actual routes. RTL simulation and generic synthesis establish a small switch baseline, without an FPGA board or ASIC implementation. The original eight communication plots remain results of the infinite-queue baseline. Physical measurements are absent and their assessment reports insufficient evidence. Trained-model quality, calibrated application throughput and energy, formal RTL liveness, and manufacturing feasibility remain unestablished.

Revision and evidence notice. This revision supersedes the original paper's numerical results and hardware-comparison claims. The original April 2026 record remains available at doi:10.5281/zenodo.19918238. That identifier refers to the historical publication; no new DOI is claimed for this revision. Baseline packet and collective timings are simulated. Prototype network cycles and transport ticks are distinct uncalibrated units. CPU numerics, RTL simulation and generic synthesis are executable evidence; capacity, endpoint-transfer and power quantities remain analytical bounds.

1 Research Question and Scope

A spherical or geodesic interconnect is an architectural idea to test against concrete workloads and physical costs. A graph can offer additional paths or shorter routes while consuming more ports, wires and energy. Neither visual symmetry nor a smaller hop count establishes a useful accelerator.

The practical question is whether a particular organization improves completed application work at a fixed accuracy, request-latency objective and implementable resource budget.

This revision distinguishes four objects and their evidence requirements. First, an implemented *flat graph model* represents 42 or 162 logical ranks connected as a triangulated sphere or a two-dimensional torus. Second, a historical Sphere-162 capacity scenario assigns hypothetical compute, memory and IO tiles to 162 positions. Third, a *new hardware proposal* places compute beside local HBM in four islands and uses a smaller global network for activations and explicit remote operations. Fourth, independent *executable prototypes* test bounded communication, numerical inference, fault semantics and local-switch RTL. The software evaluates the first object, constrains the second, and implements limited reference behaviors relevant to the third. Logical islands do not constitute a physical implementation of the proposed accelerator.

The contribution is a corrected baseline, executable reference components and a testable next design. The software supplies deterministic routes, packet-event timing, paired synthetic workloads, delivery accounting and parameter checks. The paper reports both favorable and unfavorable topology outcomes. It also derives the memory-port bottleneck that makes local data placement a more urgent architectural decision than increasing aggregate compute peaks. Prototype correctness evidence is reported separately from topology performance and physical qualification.

1.1 Corrections to the Earlier Evaluation

The earlier unloaded experiment submitted packets as a burst, allowing later probes to queue behind previous ones. The incremental scheduler reserved an entire route in packet-submission order, so a packet could claim a downstream link before a competitor that should arrive there first. The corrected unloaded path resets reservations for every probe; concurrent traffic uses chronological hop-arrival events. These changes invalidate direct reuse of the old latency distributions.

Other corrections make the comparison auditable. Sphere and torus use equal rank counts. A selected balanced coordinate partition is described as a cut estimate rather than a minimum-bisection proof. Ring and tree collectives carry the same full payload per rank. Straight express chords use Euclidean distance. Invalid routes, disconnected collectives, numeric overflow and insufficient timestamp precision are handled explicitly. Performance bounds require matching precision and account for memory-tile egress. Component power is no longer presented as full-system power or used to rank commercial accelerators.

2 Design Concepts and Reference Methods

2.1 Graph, Coordinates and Physical Package

The logical sphere recursively subdivides an icosahedron and normalizes new vertices to a common radius. At subdivision level s, its vertex and edge counts are

$$N_s = 10\,4^s + 2, \qquad E_s = 30\,4^s. \tag{1}$$

Levels one and two contain 42 and 162 vertices. Twelve vertices have degree five; the remaining vertices have degree six. The graph has no rectangular boundary, but its vertices are not all equivalent under rotation. A torus also lacks mesh boundaries, so a generic claim that spherical graphs remove a torus's corners is inappropriate.

Logical connectivity is distinct from the physical arrangement of dies. A geodesic graph might be routed on a planar package or explored in a faceted assembly. Neither follows automatically from a graph generator. A physical study must price die-edge ports, routed lengths, crossings, PHYs, HBM placement, power delivery, assembly access and cooling. Current geometry is a model input rather than an extracted package route.

2.2 Planet-Style Coordinates and Frame Reorientation

Let a rank have unit position vector $\mathbf{u}_i = (x_i, y_i, z_i)$. The proposed coordinate interface assigns latitude and longitude as

$$\varphi_i = \arcsin z_i, \qquad \lambda_i = \operatorname{atan2}(y_i, x_i). \tag{2}$$

Away from the poles, a local tangent basis is

$$\mathbf{e}_E = (-\sin\lambda, \cos\lambda, 0), \tag{3}$$

$$\mathbf{e}_N = (-\sin\varphi\cos\lambda, -\sin\varphi\sin\lambda, \cos\varphi). \tag{4}$$

A prime-meridian convention must define east at a pole. The right-handed convention above satisfies $\mathbf{e}_E \times \mathbf{e}_N = \mathbf{u}$, so a pole implementation can derive north as $\mathbf{u} \times \mathbf{e}_E$ after choosing east. These are proposed coordinate semantics; the present traffic model does not exercise a runtime coordinate interface.

A rotation $R \in SO(3)$ preserves distances and dot products:

$$(R\mathbf{u}_i)^{\mathsf{T}}(R\mathbf{u}_j) = \mathbf{u}_i^{\mathsf{T}}\mathbf{u}_j. \tag{5}$$

Changing reference-frame labels therefore neither increases bandwidth nor moves a tensor to another memory owner. Coordinates can support mapping and diagnostics, but performance improvement requires actual placement, migration or route-policy changes with their costs included. Runtime frame changes and epoch transitions remain proposed interfaces.

2.3 Reference Methods

Roofline analysis distinguishes a precision-specific compute ceiling from a bandwidth/intensity ceiling [1]. It identifies constraints; it is not an inference-runtime prediction. Here, intensity and peak rates are explicit assumptions, with no empirical utilization factor that turns the bounds into measured tokens per second.

BookSim supplies a detailed cycle-accurate network-modeling reference [2, 3]. It has not been integrated with or used to calibrate the Python baseline. The separate bounded-network prototype now includes buffering, delayed credits, arbitration and endpoint constraints. Its discrete cycles are uncalibrated, and it does not reproduce a hardware router pipeline or establish cycle-accurate chiplet performance.

Adaptive routing with an escape subnetwork requires a specified routing and channel-dependency argument [4]. The baseline uses deterministic shortest-hop breadth-first search (BFS). The bounded prototype instead uses fixed spanning-tree up/down routes with a checkable channel order. Neither implements the earlier adaptive/escape-VC proposal. The resulting fixed-route certificate is narrower than a formal hardware deadlock or liveness proof.

3 Corrected Simulation Method

3.1 Packet Events and Link Service

Each directed link has a busy-until time. A packet is routed before simulation, then scheduled one hop at a time in increasing hop-arrival time. For arrival a, previous link release b, F flits, B bits per flit and directional rate C in Gbit/s,

$$t_{\text{start}} = \max(a, b), \qquad t_{\text{end}} = t_{\text{start}} + \frac{FB}{C}\ \text{ns}. \tag{6}$$

The next arrival adds propagation and, for an intermediate router, router delay. The reference delay is four stages at 2 GHz, or 2 ns. Propagation uses an assumed 150 mm/ns wire speed. Store-and-forward timing charges the full packet's serialization at every hop. Opposite directions have

independent bandwidth, and simultaneous event ties use the input packet index. This tie rule is an explicit part of the model.

The batch API requires a fresh simulator or a reset. Incremental single-packet calls retain reservations and suit isolated or deliberately sequential operations; they must not replace batch scheduling when packets compete. A same-rank transfer records zero network latency. Unreachable destinations remain visible in delivery accounting. The simulator rejects invalid rates, packet sizes, inconsistent topologies and derived timings that overflow or lose positive durations at large absolute timestamps.

This baseline has infinite output queues. It omits input-queued switches, finite buffers, credit backpressure, VC allocation, shared endpoint injection, HBM transactions, compute and protocol overhead. Every finite trace can drain if routes remain reachable. Drain-inclusive payload rates and high-load latency curves therefore do not establish stable saturation throughput or bounded queue requirements for hardware. Section 7 describes a separate bounded-buffer implementation; none of the eight baseline plots or their tables are reinterpreted as outputs of that implementation.

3.2 Topology and Resource Budgets

The main evaluation uses 162 ranks: a level-two sphere with 480 undirected links, the same sphere with ten disjoint-endpoint antipodal chords, and a 9×18 torus with 324 links. The smaller sweep uses 42 ranks, comparing 120 sphere links against an 6×7 torus with 84 links. Surface links follow arcs at an assumed 40 mm sphere radius; chords are straight Euclidean segments. All torus links, including wraps, are assigned 4 mm. These unequal geometry assumptions remain visible.

Under `per_link`, every directional link has 256 Gbit/s; the sphere receives about 43–48% more aggregate bandwidth than torus. Under `equal_total`, sphere rates are reduced so the baseline sum of directed-link capacities matches torus. At 162 ranks, plain sphere uses 172.8 Gbit/s per direction per link instead of 256 Gbit/s. This matches one resource total, not PHY area, power, wiring cost or endpoint rate per rank. Chords consume additional ports and length rather than providing free shortcuts. Neither bandwidth policy is a complete fair-hardware comparison.

The sweep holds rank placement fixed for both graphs. Optimized placements require equal search effort and a separate experiment, especially when a workload naturally aligns with one candidate. A synthetic grouped topology exists in the source, but no radix-matched dragonfly comparison is part of the regenerated publication results.

3.3 Evaluation and Sweep Configurations

The eight general evaluation figures use 2000 paired uniform-random packets, seed 42, 162 ranks and the `equal_total` policy. Unloaded packets contain four 512-bit flits (256 bytes) and are independent probes. The load sweep uses aggregate rates of 0.001, 0.01, 0.1, 0.5, 1 and 2 packets/ns. A separate hotspot trace runs at 0.5 aggregate packets/ns. Rates are across all ranks, not per rank. The hop CDF covers all reachable source/destination pairs; latency CDFs sample the seeded packet trace.

The sensitivity study uses skewed expert dispatch and a periodic four-neighbor stencil. Two rank counts, two bandwidth policies and five paired seeds (7, 19, 42, 73 and 101) give 40 profiles. Each has 500 packets of 16 flits (1,024 bytes) at 0.1 aggregate packets/ns. Candidates receive identical rank pairs and injection times. Offered payload is 819.2 Gbit/s across all ranks. The MoE proxy excludes gating, expert compute, combine and memory access. The stencil grid directly matches torus adjacency.

Each profile also removes one seeded undirected edge in both directions before replaying traffic. The total is 160 candidate/scenario runs and 80,000 attempted deliveries. One removed link is a different failure fraction in graphs with different edge counts. These are static reachability and rerouting probes, not live fault-detection, replay or recovery tests. No unreachable packets occurred in the recorded sweep; finite-buffer drops and retransmission are outside the model.

Across seeds we report median and observed minimum–maximum paired ratios. These ranges are descriptive spreads, not confidence intervals. A median of per-seed ratios need not equal the ratio

Table 1: Corrected 162-rank evaluation under equal aggregate baseline directed bandwidth. Latencies are isolated 256-byte simulated packets. Entries are generated from `artifacts/results.json`.

Candidate	Mean hops	Max hops	p50 (ns)	p99 (ns)	p99/p50
Sphere	5.665	12	83.594	139.356	1.667
Sphere+chords	4.406	7	70.896	85.545	1.207
Torus	6.764	13	70.187	130.347	1.857

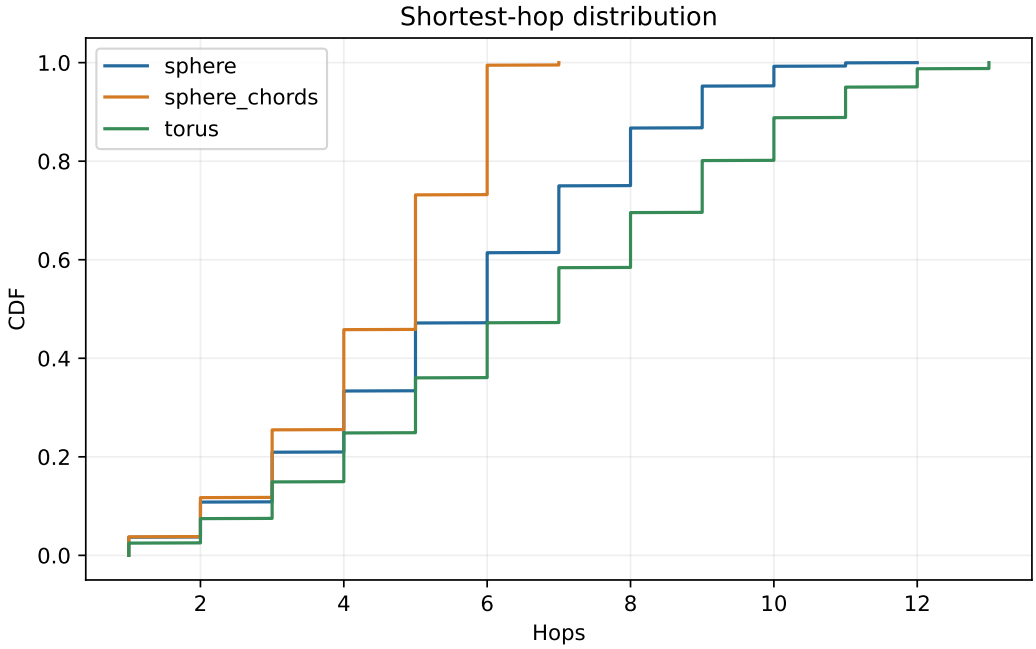


Figure 1: All-pairs shortest-hop CDF for equal-size 162-rank graphs. Chords change connectivity and resource cost. Graph distance alone does not establish latency or physical feasibility.

of median latencies. Held-out workload shapes and traces are required before claiming generality beyond these generators.

4 Regenerated Communication Results

4.1 Hop Counts and Isolated Latency

Table 1 records the corrected evaluation. Isolated latency depends on serialization and wire assumptions as well as hops. A graph with fewer hops can lose latency when links slow to match aggregate bandwidth. Figure 1 describes graph distance; Figure 2 adds the declared timing model. Neither measures a fabricated package.

Isolated p99 values are 139.356 ns for sphere, 85.545 ns for sphere with chords and 130.347 ns for torus. Corresponding p99/p50 ratios are 1.667, 1.207 and 1.857. The earlier burst-contaminated ratios are superseded. Isolated ratios describe this path distribution; they do not establish request p99 for a complete application.

4.2 Offered Load and Hotspots

Figures 3 and 7 exercise shared directed-link service under finite traces. Chronological scheduling allows a later submitted packet to reach a downstream link first when its route and injection require that order. This removes submission-order reservations that distorted the earlier congestion results. Simultaneous ties follow the documented deterministic rule.

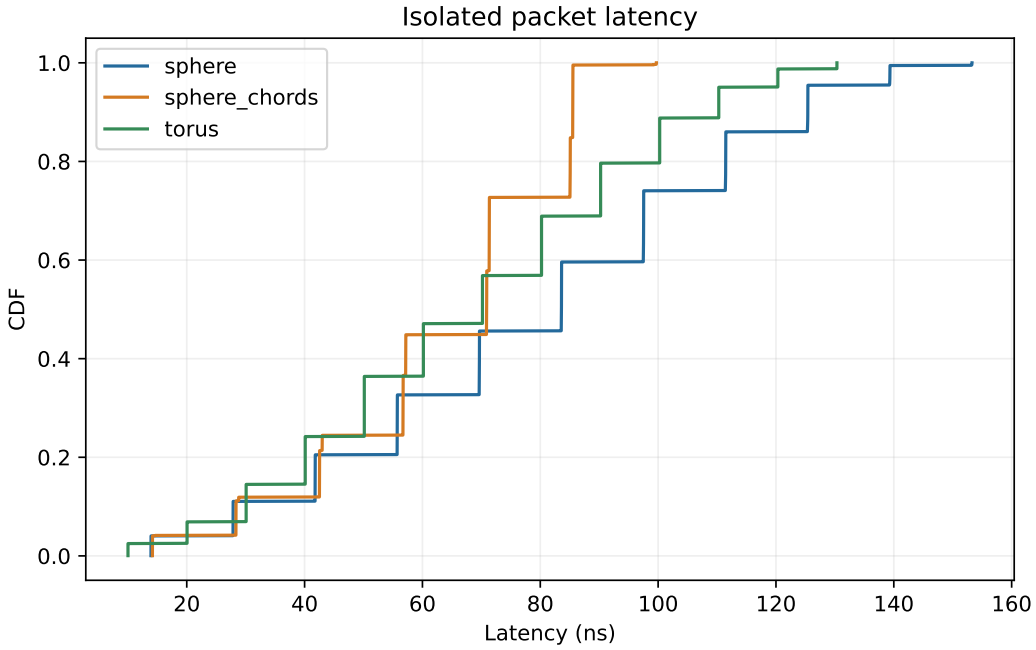


Figure 2: Corrected isolated-packet latency CDF: 2,000 paired packets, seed 42, 256 bytes each, `equal_total` bandwidth. Link reservations reset for each packet; there is no inter-probe queueing.

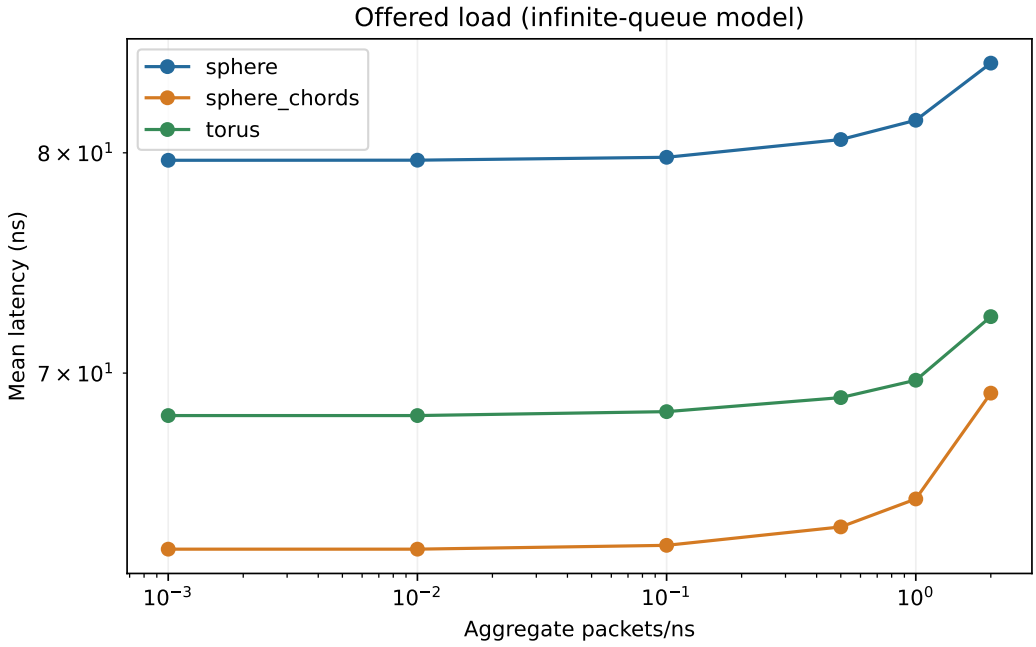


Figure 3: Mean latency versus aggregate offered packet rate in the infinite-output-queue model. The finite trace drains after injection. Curves are not hardware saturation limits or proof of bounded steady-state queues.

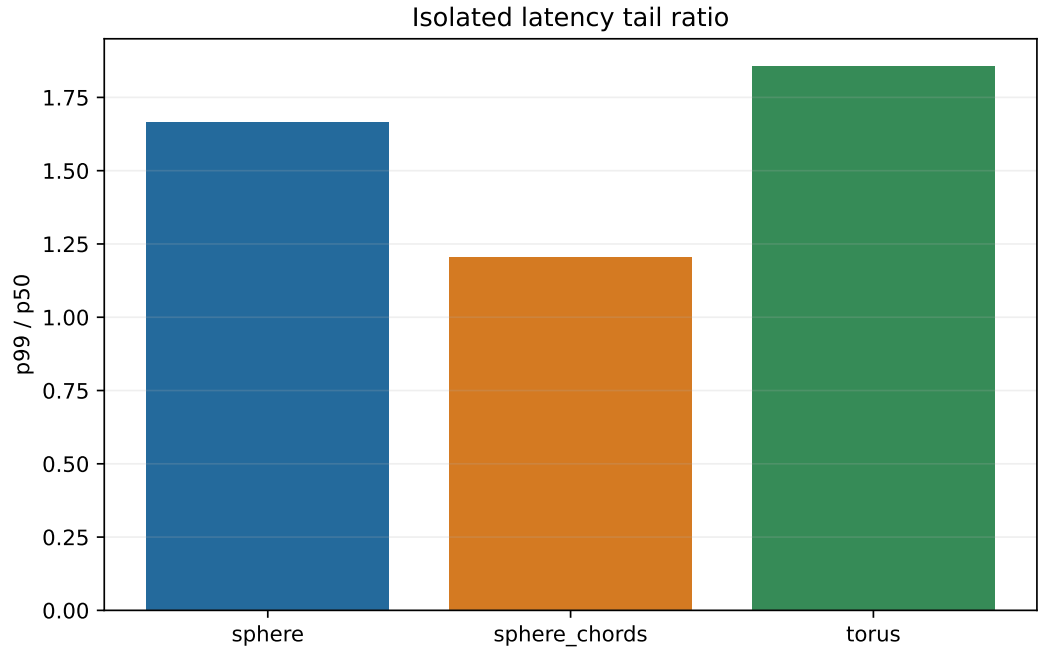


Figure 4: Regenerated isolated latency p99/p50 ratios from the same 2,000-packet trace. Ratios quantify the chosen path/timing distribution, not end-to-end inference request tails.

Table 2: Ideal matched-payload collective schedules and one selected balanced coordinate cut. Cut bandwidth is per direction, not a minimum-bisection calculation. Values come from the current JSON.

Candidate	Ring (μs)	Tree (μs)	Chosen cut (Gbit/s)
Sphere	49.803	92.21	9331.2
Sphere+chords	33.986	47.066	10494.955
Torus	11.609	67.445	4608

A hotspot concentrates demand on fewer destinations, exposing how a total budget can hide local pressure. The model has no separate endpoint injection/ejection server, memory controller or expert compute queue. A real hot expert can therefore encounter a bottleneck missing from the network-only curves. Delivery counts and complete configuration must accompany every latency result.

4.3 Collectives and the Selected Coordinate Cut

Ideal collective schedules use eight-flit ring chunks and a full $N \times 8$-flit message per tree participant. At 162 ranks the per-rank full payload is 1,296 flits, or 82944 bytes. Matching this payload corrects the earlier ring/tree mismatch. Timings omit reduction arithmetic, launch/software overhead, finite buffers and protocol costs. The schedules do not verify numerical reduction outputs. They are model experiments, not NCCL implementations or benchmarks. NCCL distinguishes algorithm from bus bandwidth, illustrating why payload and normalization must be explicit [5].

The cut procedure sorts vertices on a chosen coordinate axis and places half on either side. Crossing undirected links are counted and multiplied by the directional rate. Different balanced partitions can have different capacities. No search proves that this cut minimizes capacity. Figure 6 therefore uses "chosen balanced coordinate-cut" terminology. Neither the cut nor an arc-length sum establishes physical package routability.

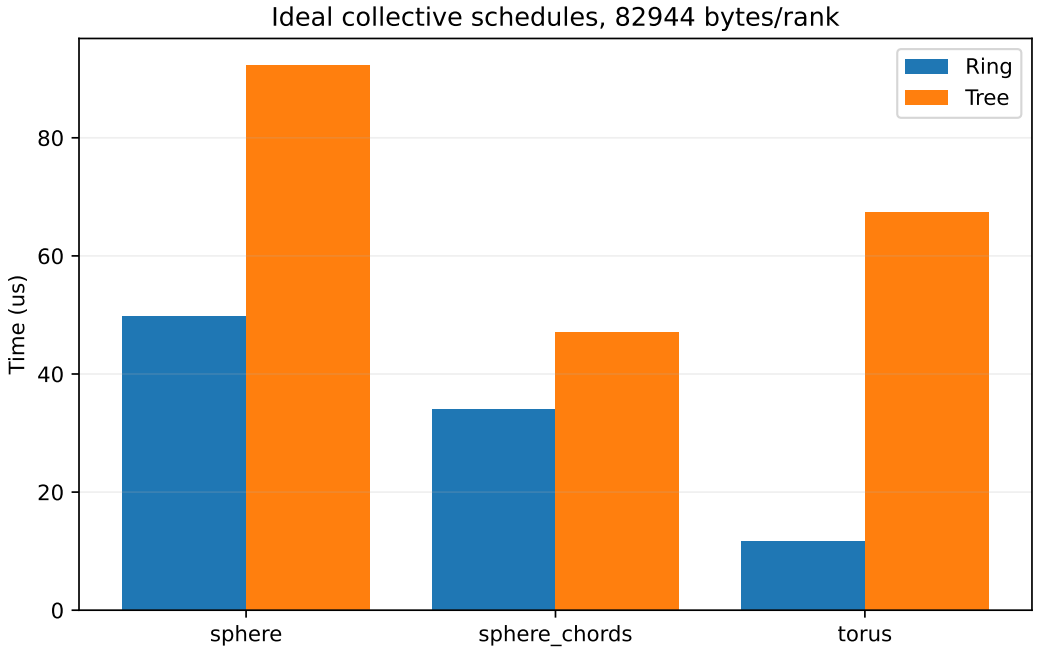


Figure 5: Ideal ring and reduce/broadcast tree schedules with the same 82,944-byte per-rank payload. Compute, software, finite-buffer transport and numerical output validation are omitted. These are not hardware all-reduce timings.

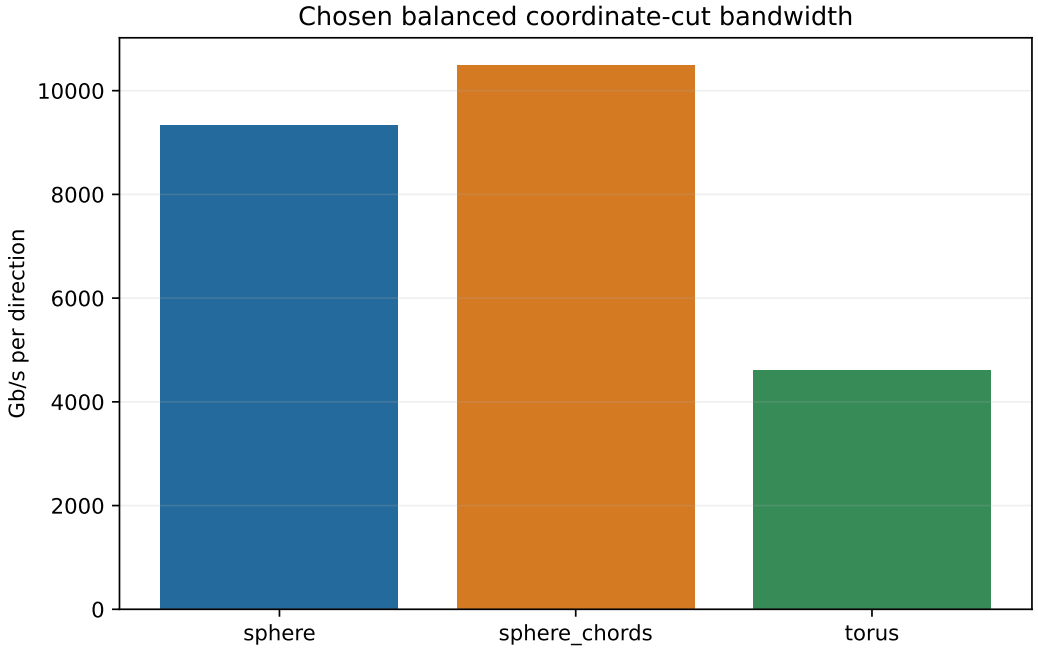


Figure 6: Capacity of one balanced coordinate-median cut per candidate under the declared directional rates. This selected-cut estimate is not a minimum-bisection proof.

Table 3: Five-seed baseline sweep. p99 columns are medians in ns. The last three columns summarize paired sphere/torus p99 ratios; greater than one means a slower sphere. Min/max are observed ranges, not confidence limits. Policies match either total directional capacity or individual link rates.

Workload	Ranks	Policy	Sphere p99	Torus p99	Ratio median	Min	Max
MoE	42	equal total	280.765	224.099	1.226	1.143	1.315
MoE	42	per link	182.895	224.099	0.834	0.776	0.914
MoE	162	equal total	512.361	442.347	1.158	1.119	1.326
MoE	162	per link	344.777	442.347	0.779	0.771	0.879
Stencil	42	equal total	287.171	34.027	8.44	8.44	8.859
Stencil	42	per link	204.886	34.027	6.021	6.021	6.021
Stencil	162	equal total	542.321	34.027	15.938	15.199	17.453
Stencil	162	per link	363.258	34.027	10.676	10.037	12.018

4.4 Workload Sensitivity and Static Failure Probes

Table 3 presents every baseline cohort in the five-seed sweep. At equal total bandwidth, sphere has higher MoE p99 at both rank counts. Equal per-link rates lower sphere p99 while supplying approximately 1.429 times torus aggregate bandwidth at 42 ranks and 1.481 times at 162 ranks. The resource policy changes the conclusion.

The stencil favors torus by construction: neighbors in its periodic rank grid are physical torus neighbors. Sphere uses the same rank IDs without a placement search. Large stencil ratios identify a workload/mapping control, not a universal outcome for scientific computing. A geodesic-domain or unstructured-grid workload needs its own trace and equivalent placement optimization for every candidate.

Static single-edge removal left sampled graphs connected. Most p99 ratios stayed near one, consistent with rerouted traffic affecting too few packets to move the 99th percentile; affected-route fractions were not measured separately. A sampled 42-rank stencil torus case tripled p99. Some MoE cases had slightly lower latency after removal because deterministic BFS redistributed routes. This is not evidence that physical failures are beneficial. The JSON records each failed edge, delivery count and scenario so these effects can be inspected.

4.5 Energy Proxy

Figure 8 retains only an explicitly unvalidated path-count proxy. It assumes 1.5 pJ per link-bit and 0.5 pJ per router-bit at every rate and length, with an additional 0.5 pJ/bit term in the reported formula. These constants are not calibrated PHY measurements. The proxy omits static power, packet/control overhead, memory, compute, regulators, host and cooling. It cannot establish energy per token, PFLOPS/kW or superiority over a commercial accelerator.

5 Capacity Bounds and the Locality Constraint

5.1 Historical Reference Assumptions

The analytical scenario retains 113 compute tiles, 25 memory-controller tiles and 24 cache/IO tiles to expose the original assumptions. Each memory tile has one assumed 36 GB stack at 1.2 TB/s locally. The sum is 900 GB and 30 TB/s at the stacks. Neither quantity establishes uniform latency, coherent sharing, feasible stack placement or simultaneous remotely achievable bandwidth.

Each memory tile has an optimistic six-port outward budget at 256 Gbit/s per direction. A port

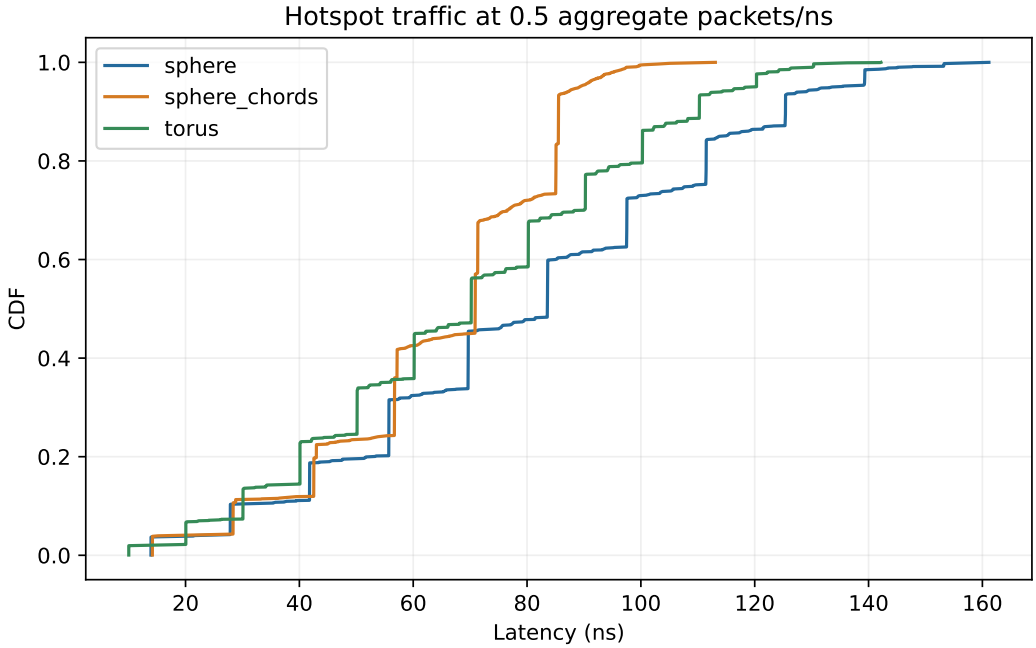


Figure 7: Synthetic hotspot latency at 0.5 aggregate packets/ns. Directed-link contention is modeled; shared endpoint injection and memory-controller service are omitted and could constrain a real hot destination.

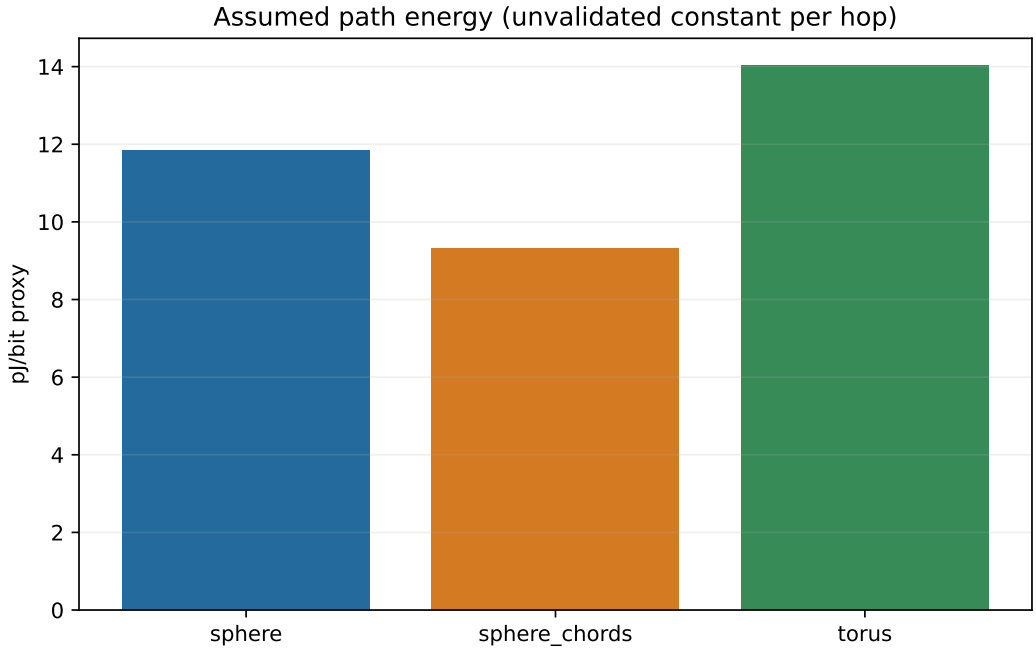


Figure 8: Assumed path-energy proxy from hop count and fixed unvalidated per-bit constants. It is neither a PHY measurement nor a full-system energy or efficiency comparison.

Table 4: Assumption-driven bounds regenerated from `artifacts/performance_model.json`. Full-system power is unknown; summed stack capacity does not imply uniform remote service.

Quantity	Value
Assumed dense FP8 peak (PFLOPS)	56.5
Assumed HBM capacity (GB)	900
Assumed summed local HBM bandwidth (TB/s)	30
Remote HBM endpoint ceiling (TB/s)	4.8
Necessary local-byte fraction (%)	84
Partial component power assumption (W)	2373.624
Full-system power/efficiency	Not established

supplies 32 GB/s; six provide 192 GB/s. Summing over 25 memory tiles gives

$$B_{\text{remote}} \leq 25 \times 6 \times \frac{256}{8}\ \text{GB/s} = 4.8\ \text{TB/s}. \tag{7}$$

Cuts, protocol overhead, controller arbitration, placement and contention can tighten this endpoint ceiling. Not every sphere vertex has six surface links; the uniform six-port budget is deliberately optimistic. Routing alone cannot remove this bound.

Consuming the full assumed stack aggregate while remote demand stays below that ceiling needs a local byte fraction of at least

$$f_{\text{local}} \geq 1 - \frac{4.8}{30} = 84\%. \tag{8}$$

This is necessary, not sufficient. A local access path, controller service, compute consumption and realizable package must still exist.

5.2 Compute, Precision and Power

Assuming 500 dense FP8 TFLOPS per compute tile sums to 56.5 PFLOPS. No synthesized engine or measured kernel establishes that peak. A single 256×256 array at 2 GHz, issuing one FMA per cell per cycle and counting two FLOPs per FMA, supplies 262.144 TFLOPS. Array count, formats, accumulation, data supply and implementation power must be specified before accepting a different peak. FP8 cannot substitute for BF16, FP16 or FP64 work, and sparse rates cannot silently become dense rates.

For intensity I_p at precision p, the revised capacity code reports

$$P_p \leq \min(P_{p,\text{compute}}, B_{\text{memory}} I_p). \tag{9}$$

Precision must have a matching compute peak. Decimal TB/s times FLOP/byte gives TFLOPS. This does not model usable memory fit, tensor placement, kernel utilization, overlap, host overhead, collectives or numerical error. It is a constraint, not a workload benchmark or justification for an arbitrary spillover penalty.

The power model reports 2373.624 W as a component subtotal under declared tile and dynamic-link assumptions. HBM stacks, static fabric, external IO/host, power conversion and cooling are missing. Full-system power is explicitly unknown. Dividing an assumed peak by this subtotal would overstate the evidence; no efficiency ranking is reported.

6 Proposed Compute-Near-HBM Accelerator

6.1 Four-Island Planning Design

The next architecture study starts with four islands. Each has two compute chiplets, one local HBM stack, local SRAM and a memory/IO complex. Eight compute chiplets and four stacks define a

study size, not a purchased component list or routed package. A wide local path serves resident weights and state. A separate hierarchical inter-island network carries activations, collectives and explicitly budgeted remote operations.

Expert weights stay beside their compute; attention/KV shards have stable owners. The runtime exposes NUMA locality and admits work against the most constrained island rather than treating aggregate free bytes as immediately available to every rank. A geodesic graph, short-wire mesh/torus and switched hierarchy remain candidates. Selection needs the same application, local-memory budget and priced port/wiring constraints. The flat-graph experiments do not test this hierarchy. The new numerical prototype implements four logical owners and a small tree network; it does not implement separate compute devices, HBM controllers or a physical hierarchical package.

Planar construction is the initial package study. A physical faceted sphere remains an optional mechanical/cooling experiment. The level-two, radius-40 mm graph has approximately 11–13 mm surface-neighbor arcs, not the earlier claimed 4–7 mm. Logical arcs do not establish PHY reach, die escape routing or equal per-bit energy. Package construction and connectivity must be evaluated separately.

6.2 Runtime and End-to-End Workload Contract

Memory-resident MoE inference supplies a concrete lead application. A full trace must include gating, dispatch, expert computation, combine, attention/KV access, tensor-parallel phases and host scheduling. DeepEP provides a reference for expert-parallel dispatch/combine on its supported platforms [6]; it is not a backend for this hypothetical device.

An execution generation should identify weights, tensor ownership, communicator participants and routes. Dispatch carries token/expert identity, bounds and generation. The receiver checks messages before expert execution. Combine must return each expected contribution exactly once and preserve numerical correctness. A timeout or reset must account for every outstanding operation and complete or explicitly abort its generation. The transport and inference prototypes exercise a bounded subset of these requirements in software. Static edge removal itself establishes none of these live protocol properties.

Tiny-packet latency is not layer latency. With 128 tokens, width 8,192, top-$k = 2$ and two-byte activations, all-remote assignments require at most 4 MiB of dispatch payload. Returning equal-width outputs adds 4 MiB. An ideal 192 GB/s endpoint alone requires about 43.7 μs for the sequential payload transfers, before metadata, queueing, compute and software. A full byte ledger is required before converting network plots into tokens/s.

6.3 Transport, Control and Frame Semantics

Hardware transport needs bounded buffers, credit conservation, CRC/replay, ordering, timeouts, error containment and reset semantics. Adaptive routing needs legal VC transitions and a verified escape dependency graph including request/response and injection/ejection dependencies. Fault and epoch transitions must preserve the relevant properties. There is no claim that three VCs suffice without those specifications and proofs. The reference components in Section 7 test finite queues and bounded recovery separately; they are not an integrated adaptive hardware transport.

Management should remain available during data-network degradation and control safe power/clock sequencing. Physical IDs and memory ownership should stay stable under coordinate reorientation. A prepare/commit scheme needs tests with old/new-generation traffic in flight, rollback and partial management failure. The original sub-microsecond frame-transition claim is withdrawn because the protocol is not modeled.

Table 5: Generated implementation status for v1.2.0. Passing software/RTL checks does not close the physical or application-value gates.

Component	Executed evidence and boundary
Finite-buffer model	Credit conservation, fixed-route DAG and bounded cutoff verified
Tiny inference	Independent CPU numerical reference and contribution accounting pass
Transport reference	Recovery, duplicate suppression, abort and generation cases pass
RTL prototype	Simulation and generic synthesis pass; no FPGA board result
Physical validation	Harness implemented; no hardware measurements supplied

Table 6: Selected generated prototype results. Network cycles include credit drain; protocol ticks and RTL clock cycles are separate units. Generic cells are synthesis statistics, not physical area.

Quantity	Result
Bounded-network baseline packets delivered	24
Bounded-network elapsed model cycles	41
Numerically processed prompt and generated tokens	8
Completed weighted expert contributions	16
Maximum absolute cached/reference logit error	`6.66e-16`
Executed RTL fabric configurations	4
Delivered RTL words across those configurations	17188
Default word-fabric generic synthesized cells	4913

6.4 Physical, Electrical and Thermal Evidence

PHY selection needs actual channel construction, lengths, bends, connectors, layers and return paths. SI/PI analysis must cover the selected IP's requirements across process, voltage and temperature. An arbitrary eight-lane rate in Python does not establish compliance with an interface standard or feasibility on a spherical shell.

Thermal acceptance needs local junction limits and workload hotspots, not average shell heat flux. A planar cold plate and any later segmented shell should use consistent inlet conditions, flow, allowable temperatures and measured wall power. Heater vehicles can test contact pressure, interfaces, pressure drop, leaks and fault responses before functional silicon. No thermal qualification, yield, serviceability assessment or silicon signoff has been completed.

7 Executable Prototype Implementations and Evidence

Version v1.2.0 adds independent implementation evidence dated 2026-11-10. Table 5 summarizes the generated status, while Table 6 reports selected executed counts. The source is `artifacts/prototype_validation.json`; its manifest identifies the source, configuration and test bytes. These results do not replace the infinite-output-queue sphere/torus experiments or calibrate their nanosecond timing assumptions.

7.1 Bounded Flit Queues, Credits and Fixed Routes

`sim/finite_buffer.py` has a finite FIFO for each physical input and VC, explicit flits in flight, delayed credit return, bounded endpoint injection/ejection and sink backpressure. Each physical input supplies at most one flit per cycle across its VCs, and each directed link sends at most one. Only FIFO heads compete. Deterministic round-robin arbitration and positive router/link delays make contention and head-of-line blocking observable. Per-link/VC conservation requires

$$c_{\text{available}} + n_{\text{resident}} + n_{\text{flight}} + n_{\text{return}} = c_{\text{capacity}}. \tag{10}$$

The model checks this identity and exact flit residency after every cycle. Completion requires all accepted packets delivered, all buffers and link events empty, and every credit returned. A maximum-cycle cutoff preserves pending and uninjected flits in its report; disconnected requests are explicitly rejected. A completed subset is never silently reported as full delivery.

Routes follow a deterministic spanning forest, up to the least common ancestor and then down to the destination. Non-tree links carry no traffic. For maximum forest depth H, a child at depth d has upward channel rank $H - d$ and downward rank $H + d$. Every consecutive channel dependency on an allowed route strictly increases in rank. The generated certificate enumerates the actual channels and dependencies, giving a checkable acyclic order for this fixed policy. Packets retain their initial VC. This is not adaptive routing, wormhole header/body allocation, a VC-transition proof, a formal RTL proof or a progress guarantee under a permanently stalled sink.

The executed scenarios include an ordinary drain, a finite sink pause, an indefinitely paused sink stopped at the configured cutoff, and an unreachable cross-component request. Independent trace replay verifies FIFO actions, physical-port limits and exact flit/credit accounting; a separate graph traversal checks the reported dependency graph. No clock period, PHY bandwidth or hardware latency is inferred from these logical cycles.

7.2 Actual Tiny Causal MoE Inference

`sim/moe_inference.py` executes token and positional embeddings, normalization, multi-head causal attention, a bounded KV cache, stable top-k routing, expert MLPs, weighted combination, vocabulary logits and greedy generation in NumPy float64. The checked configuration uses a small synthetic model, not pretrained weights or a language-quality benchmark. An independent full-prefix implementation recomputes vectorized causal attention and all expert intermediates without using the cached decoder, its schedule or its numerical helpers. The comparison checks every prefill logit, logits after each generated token, and matching greedy tokens at absolute and relative tolerances of 10^{-10}.

All arithmetic runs in one CPU process. Four logical islands own resident expert views, with expert e assigned to island $e \bmod 4$. Island zero owns attention, embeddings, the head and the full KV cache; the KV cache is not distributed across devices. Dispatch and combine use fixed tree links 0–1, 0–2 and 2–3. Actual activation/output bytes are staged on the host and made available to their logical destination only after the finite-buffer model validates each required packet. Local experts bypass the network. Each phase drains its credits before dependent work proceeds; the next token waits for complete combination and cache commit.

A per-owner inventory counts actual NumPy array sizes for resident weights, KV arrays and reserved workspaces. Admission enforces those declared tensor budgets. The inventory excludes Python overhead, BLAS temporaries, staged payloads and reference computation, so it is not a bound on process peak memory or evidence of physical device allocation. Separate byte ledgers record payload, flit padding, hop-weighted traffic, local bypass, expert-weight reads and KV reads/writes. Configurable operation-count proxies supply assumed compute cycles. Neither those cycles nor the CPU calculation establish accelerator tokens/s, trained-model accuracy, energy or multidevice speedup.

7.3 Payload Recovery and Uncertain Outcomes

`sim/transport.py` carries actual immutable payload bytes in bounded DATA/ACK sessions. CRC32 covers identity and payload; generation, request, sequence and endpoint checks reject mismatches. Duplicate suppression accepts each admitted contribution once, while retransmitted copies can elicit a new ACK. Retries, request/event capacity and reservations have explicit bounds. Retry exhaustion or a generation barrier terminates old work and releases its reservation exactly once. The shared sender/receiver admission oracle and retained deduplication state are explicit reference-model assumptions.

The executed recovery cases inject drops, corruption, duplication and reordering, plus partition, all-ACK-loss and generation-transition cases. An additional inference experiment transports three actual remote expert output vectors, alongside one local output, under DATA loss, corruption, duplication and retry-ACK loss. Unique acknowledged outputs combine once and match the independent numerical reference. When all ACKs are lost, receiver acceptance can occur even though the sender ultimately aborts. The decoder then refuses combination and cache commit; it does not claim that an abort rolled back the receiver or established global exactly-once execution.

Transport ticks are separate from bounded-network cycles and RTL clock cycles. The implementation provides neither authenticated messaging nor durable recovery after endpoint restart. Reconciliation of uncertain application effects, distributed commit and a refinement into hardware transport remain open integration work.

7.4 Local Word-Fabric RTL and Physical Evidence Gate

`rtl/word_fabric_top.sv` and `rtl/sync_fifo.sv` implement a single-clock word switch with finite ingress FIFOs, ready/valid backpressure, round-robin output service and identity metadata. A stalled output retains its data and metadata; synchronous reset explicitly cancels queued words. Actual Icarus Verilog simulations compare cycle-level outputs with an independent Python oracle, including contention, stalls, pointer and sequence wrap, reset while full, and non-power-of-two parameters. Yosys runs generic flattened synthesis and structural checks. The evidence records tool versions, source/trace hashes, transfer counts and generic cells.

This RTL is a local word-switch baseline, not a bit-for-bit realization of the Python multi-hop flit network or reliable protocol. Its generic cell count is not ASIC area, FPGA LUT usage, frequency, power or timing closure. No target board, pin constraints, routed bitstream, FPGA bring-up or measured hardware result is supplied. Formal liveness, CDC/RDC, PHYs, HBM and full accelerator integration remain outside this implementation.

The physical measurement harness loads declared records and checks their provenance, calibration metadata, configured lane and corner limits, and required coverage. It computes one-sided BER confidence bounds and checks temperature, power, voltage and coolant samples. The bounds assume independent bit trials; instrument uncertainty is recorded without automatic guard-banding, and provenance authenticity needs independent review. Missing, malformed or synthetic inputs cannot qualify as physical evidence. The supplied missing-hardware scenario reports `INSUFFICIENT_EVIDENCE`. No hardware measurements were supplied, so this gate remains open; the harness itself is not compliance certification.

8 Validation, Reproduction and Release Integrity

8.1 Software and Experiment Checks

Regressions cover hand-derived timing, chronological competition, full-duplex independence, disconnected inputs, graph invariants, geometry, workload accounting, precision checks and numeric adversarial cases. CI runs on Python 3.10 and 3.12. Additional checks exercise bounded-network invariants, independent numerical inference, actual contribution recovery and RTL contracts. They verify the scoped implementations rather than hardware application performance or energy. Dedicated RTL validation requires actual simulator and synthesis tools; an absent tool cannot count as executed RTL evidence.

The publication build regenerates evaluation and capacity JSON, the paired-seed sweep, eight figure pairs and this paper's numeric macros/table rows. A separate prototype runner executes the bounded-network, inference, fault, transport and actual-tool RTL scenarios and generates its own JSON, macros and source/output manifest. Tables consume `paper/generated_results.tex` and `paper/prototype_results.tex`, not manually transcribed JSON values. Manifests record environment, configuration, source fingerprints and output hashes. Prototype test files are identified by

hash; the experiment runner does not substitute for separately executing the unit suite. Fingerprints identify the exact files used even if a working tree was not committed at calculation time. The separate paper-build manifest associates the PDF with its TeX inputs and figure files.

Reference commands, from the repository root, are:

```
python -m pip install -r requirements.txt
python -m unittest discover -s tests -v
python scripts/rebuild_publication.py
python scripts/run_prototype_validation.py
python scripts/run_prototype_validation.py -check
python scripts/build_paper.py
python scripts/build_paper.py -check
```

The paper builder runs LaTeX from the correct working directory and checks its recorded inputs. Dependency versions can change percentile rounding or rendering. In a fixed numerical environment, repeated JSON should agree; source and output hashes make disagreements inspectable. Determinism is conditional on the recorded environment, not a promise across every Python, NumPy, Matplotlib or TeX version.

8.2 Progression Beyond the Model

The next revision should follow the dominant measured bottleneck and a frozen decision rule. The validation plan proposes at least 15% more useful application throughput per joule at the same accuracy and service target, with at most 5% p99 regression in required held-out cases. These are proposed gates, not achieved results. Estimated power supports screening, not proof of product efficiency. Revise goals openly before inspecting held-out results when requirements change.

A practical sequence is:

1. Freeze workload shapes, precision/accuracy, budgets and held-out traces. Retain losing cases and failed runs alongside wins.
2. Extend the tiny numerical reference to trained models and calibrated compute/HBM service, broader workload traces and placement sensitivity. Retain complete byte accounting and independently checked outputs.
3. Cross-check the bounded network against detailed router models and refine transport into RTL with formal ordering, credit, liveness and fault-transition assertions under explicit assumptions.
4. Integrate the local word-fabric baseline into a target-board FPGA demonstrator with constraints, trace I/O and independent scoreboards. Measure bring-up; FPGA area/power is not ASIC package PPA.
5. Measure representative channels, power-delivery transients, thermal interfaces and fault responses before committing to a package.
6. Advance only with application, timing, CDC/RDC, PPA, DFT, SI/PI, thermal, yield and production-test evidence for every requirement.

Each iteration needs source/config revision, raw per-run data, environment, owner and acceptance decision. Statistical inference should resample independent runs or trace windows rather than treat correlated packets as independent. Live recovery requires corruption, duplication, timeout and generation tests beyond connectivity.

8.3 Publication History and Availability

Source is available at https://github.com/igenius0/sphere-chiplet-fabric. This revision synchronizes the paper, figures, results and citation metadata with the corrected baseline and separate

executable prototypes. The April bundle remains recoverable through the original Git tags and DOI. Its old performance and efficiency claims are superseded and must not be reused as validated measurements. A future DOI deposit must identify its actual released version; the historical identifier is not relabeled as a deposit of this revision.

Software is MIT-licensed; paper text is CC BY 4.0. Citation metadata identifies the current revision and preserves the historical record as prior work. Documentation separates implemented software, observed simulation outputs, analytical assumptions and planned hardware validation.

9 Conclusion

Corrected experiments make the spherical-interconnect hypothesis testable without presuming its outcome. The flat sphere loses tested synthetic MoE p99 comparisons when aggregate bandwidth is matched, and wins when equal link rates also supply more aggregate bandwidth. The stencil exposes placement effects. Static failed edges probe route sensitivity, not live recovery. These findings replace universal topology and efficiency claims.

The strongest constraint is locality: 30 TB/s of assumed stack bandwidth faces an optimistic 4.8 TB/s remote memory-port ceiling. Four compute/HBM islands with explicit ownership and a hierarchy provide a concrete next study. Geodesic connectivity and physical spherical packaging remain separate choices. The synchronized publication and software supply a reproducible baseline; application benefit and physical feasibility remain questions for the next measured iteration. The new bounded-network, actual numerical inference, recovery and local RTL components make selected behaviors executable and independently checkable. They do not turn logical cycles into calibrated performance, fixed-route acyclicity into a formal accelerator proof, or missing measurements into a physical validation result.

References

[1] S. Williams, A. Waterman, and D. Patterson, "Roofline: an insightful visual performance model for multicore architectures," *Communications of the ACM*, vol. 52, no. 4, pp. 65–76, 2009. https://doi.org/10.1145/1498765.1498785.

[2] N. Jiang, D. U. Becker, G. Michelogiannakis, J. Balfour, B. Towles, J. Kim, and W. J. Dally, "A detailed and flexible cycle-accurate network-on-chip simulator," in *Proc. IEEE ISPASS*, 2013, pp. 86–96. https://doi.org/10.1109/ISPASS.2013.6557149.

[3] BookSim developers, "BookSim 2.0," official source repository. https://github.com/booksim/booksim2. Accessed 9 November 2026.

[4] J. Duato, "A new theory of deadlock-free adaptive routing in wormhole networks," *IEEE Transactions on Parallel and Distributed Systems*, vol. 4, no. 12, pp. 1320–1331, 1993. https://www.csl.cornell.edu/courses/ece5750/duato.tpds93.pdf.

[5] NVIDIA, "Performance reported by NCCL tests," official documentation. https://github.com/NVIDIA/nccl-tests/blob/master/doc/PERFORMANCE.md. Accessed 9 November 2026.

[6] DeepSeek-AI, "DeepEP: an efficient expert-parallel communication library," official source repository. https://github.com/deepseek-ai/DeepEP. Accessed 9 November 2026.